

Theeraj Chandra

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EDUCATION

University of Waterloo

Sep 2022 – Apr 2027

Bachelor of Applied Sciences in Computer Engineering

- **Relevant courses:** Real-time OS, Databases, Computer Networks, Systems Programming and Concurrency

TECHNICAL SKILLS

Languages: Rust, Python, Ruby, C, C++, Go, Java, GraphQL, TypeScript, Verilog

Libraries & Frameworks: Ruby on Rails, React, PyTorch, Angular, Next.js

Tools & Databases: Buildkite, Artifactory, Docker, Sentry, Redis, PostgreSQL

EXPERIENCE

Software Engineer Intern (QE Infra) | *TypeScript, GraphQL, Go, Buildkite*

Jan 2026 - Apr 2026

StackAdapt

Toronto, ON

- Optimized high-throughput analytical queries for a 14GB PostgreSQL dataset, achieving a **42x performance speedup** (5m to 7s) by re-architecting composite indices and resolving data non-determinism caused by timestamp collisions.
- Authored the design specification for an **AI-powered Triage Engine** utilizing a headless MCP architecture and temporal knowledge graphs to automate failure root-cause analysis across distributed CI pipelines.
- Engineered a high-concurrency **TypeScript Service** that scrapes historical data from 500+ Buildkite streams; built this data-aggregator to feed LLMs the context needed to train them to accurately distinguish between real bugs and environmental flakiness.
- Optimized CI throughput by building a custom **workload scheduler** in **Go** to dynamically shard test suites across distributed clusters; reduced the feedback loop for our E2E pipelines from 40 minutes to **under 8 minutes**.

Software Engineer Intern | *React.js, TypeScript, Material UI*

Jan 2025 – Apr 2025

SuperWorld Inc.

Miami, FL

- Engineered a geolocation-based **virtual check-in** feature for SuperWorld Map using the Haversine formula algorithm, enabling users to authenticate location proximity within a 300m radius, thus enhancing user engagement through location verification
- Played a key role in developing our proprietary **conversational AI interface** powered by Meta LLaMA 3, enhancing user retention through personalized map interactions, and intuitive mobile-first UX
- Built a React + TypeScript dashboard with async parallel API calls for real-time KPI updates and interactive Recharts visualizations

Test Engineer Intern | *Azure DevOps, Perl, Postman*

Jan 2024 – Aug 2024

First National Financial LP

Toronto, ON

- Developed **Python** and **Perl** scripts to automate pooling target validation and improve testing processes, ensuring higher reliability and reducing manual efforts by **30%**
- Tested mortgage broker tools for BMO, TD, and Manulife on staging and QA environments, performing **regression**, **UAT**, and **exploratory** tests to validate functionality and client-specific requirements

PROJECTS

Zero-Allocation Limit Order Book | *Rust*

- Engineered a matching engine from scratch in Rust achieving **160ns** average latency and **6.2M ops/sec**, outperforming typical production exchange requirements by an order of magnitude.
- Architected a **slab/arena** allocator and intrusive data structures, eliminating heap churn and lock contention to ensure deterministic $O(1)$ allocation and cache-friendly $O(\log P)$ matching.

RTOS Kernel | *C, ARM Assembly*

- Developed a preemptive RTOS kernel from scratch for ARM Cortex-M4, supporting 16 concurrent tasks with EDF scheduling algorithm
- Implemented kernel primitives including SVC-based system calls, PendSV/SysTick interrupt handlers, and a custom heap allocator with block coalescing that reduced memory fragmentation by 40%+